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## RESEARCH ARTICLE

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### Key Points:

- Canopy temperature variation is closely related to plant hydraulic traits that govern stomatal regulation
- Sensitivity of canopy temperature to changing air temperature and vapor pressure deficit differs significantly among species
- Difference in canopy temperature among species is comparable to the surface temperature variation caused by land cover change

### Supporting Information:

- Supporting Information S1

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## High Heterogeneity in Canopy Temperature Among Co-occurring Tree Species in a Temperate Forest

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**Abstract** Trees regulate canopy temperature ( $T_c$ ) via transpiration to maintain an optimal temperature range. In diverse forests such as those of the eastern United States, the sensitivity of  $T_c$  to changing environmental conditions may differ across species, reflecting wide variability in hydraulic traits. However, these links are not well understood in mature forests, where  $T_c$  data have historically been difficult to obtain. Recent advancement of thermal imaging cameras (TICs) enables  $T_c$  measurement of previously inaccessible tall trees. By leveraging TIC and sap flux measurements, we investigated how co-occurring trees (*Quercus alba*, *Q. falcata*, and *Pinus virginiana*) change their  $T_c$  and vapor pressure deficit near the canopy surface ( $VPD_c$ ) in response to changing air temperature ( $T_a$ ) and atmospheric VPD ( $VPD_a$ ). We found a weaker cooling effect for the species that most strongly regulates stomatal function during dry conditions (isohydric; *P. virginiana*). Specifically, the pine had higher  $T_c$  (up to 1.3°C) and  $VPD_c$  (up to 0.3 kPa) in the afternoon and smaller sensitivity of both  $\Delta T (=T_c - T_a)$  and  $\Delta VPD (=VPD_c - VPD_a)$  to changing conditions. Furthermore, significant differences in  $T_c$  and  $VPD_c$  between sunlit and shaded portions of a canopy implied a non-evaporative effect on  $T_c$  regulation. Specifically,  $T_c$  was more homogeneous within the pine canopy, reflecting differences in leaf morphology that allow higher canopy transmittance of solar radiation. The variability of  $T_c$  among species (up to 1.3°C) was comparable to the previously reported differences in surface temperature across land cover types (1°C to 2°C), implying the potential for significant impact of species composition change on local/regional surface temperature.

## 1. Introduction

One of the important roles of forests in the climate system is the modulation of energy and water exchanges between land and the atmosphere, which can modify surface temperature and influence local/regional air temperature (Alkama & Cescatti, 2016; Novick & Katul, 2020; Zhao & Jackson, 2014). Recently, substantial efforts have been made to improve understanding of this process in order to clarify the direction and sensitivity of surface temperature change, which will enable a more holistic perspective on the climate mitigation potential of forests. At spatial scales larger than stand/ecosystem scale, the surface temperature is controlled by surface-atmosphere feedbacks (Monteith & Unsworth, 2014). At finer scales (i.e., individual tree scale), canopy temperature ( $T_c$ ), which is the surface temperature for an individual tree, is a result of the energy balance between net radiation, latent heat flux, and sensible heat flux. These processes are controlled by environmental factors including air temperature, humidity, wind speed, canopy structure, and evaporative heat loss regulated by plants (i.e., transpiration; Jones, 2004). Given that changes in the canopy structure take place at a relatively long time scale (weeks to months), the sensitivity of  $T_c$  is largely dependent on shorter-term changes in environmental conditions and associated plant water regulation (hours to days; Mahan & Upchurch, 1988).

The sensitivity of  $T_c$  to changing conditions can vary across species due to the wide range of different hydraulic traits that influence stomatal regulation (e.g., from isohydric to anisohydric; McDowell et al., 2008; Yi et al., 2019). Species-level differences in stomatal conductance, in turn, cause different evaporative cooling effects (i.e., heat loss at the leaf surface during transpiration due to the water phase transition from liquid to vapor) among species. Improved understanding of the species-specific sensitivity of  $T_c$  is especially important for areas such as the eastern United States, where tree species composition is highly dynamic. Specifically, in this region, the combined influence of fire suppression, management, and long-term

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climate changes in the twentieth century has led to pronounced shifts in species composition in deciduous forests—from forests dominated by more anisohydric oak species, to forests dominated by more isohydric, non-oak species (Abrams, 2003; Fei et al., 2011; Flatley et al., 2013).

The balance of  $T_c$  with changing environmental conditions is also important to the plant itself because virtually all plant functions are sensitive to temperature. Temperature outside the optimal range can reduce carbon assimilation, water regulation, and other metabolic processes, acting as a stress (Berry & Bjorkman, 1980). Differences in  $T_c$  among species further imply different evaporative gradients from the leaf to air (vapor pressure deficit, VPD) across species by influencing saturation vapor pressure inside the leaf (Helliker & Richter, 2008), with elevated VPD acting as water stress near the leaf surface (Grossiord et al., 2020). Investigating how hydraulic traits mediate temperature and VPD stress, and the consequences for  $T_c$ , will improve the understanding of plant vulnerability to predicted changing climate in the future, which is expected to be warmer and drier in many places (Ficklin & Novick, 2017; Intergovernmental Panel on Climate Change [IPCC], 2018; Novick et al., 2016).

Although there have been a few studies investigating tree  $T_c$  across species (e.g., Gimenez et al., 2019; Leuzinger & Körner, 2007; Leuzinger et al., 2010), we still don't know if shifts in hydraulic trait diversity affect  $T_c$  in ways that are meaningful for physiological function and local micro-climate. One of the major causes of lacuna is the lack of continuous, high-frequency, long-term monitoring of both  $T_c$  and plant water use together. This has been challenging until the recent advance of thermal imaging cameras, or TICs, which can resolve the thermal signature of individual trees. The  $T_c$  data in most previous studies utilizing TICs were not continuously collected as part of a long-term monitoring protocol; images were taken by choosing a number of times and days of favorable conditions for measurements (clearness, wind, temperature, etc.), which may only represent limited “normal” conditions. Long-term, continuous observations of  $T_c$  over a wider range of environmental conditions provide a valuable opportunity to understand links between canopy temperature and physiological response to changing environmental conditions. Infrared radiometers are widely used for continuous, stand-scale  $T_c$  measurements (Blonquist et al., 2009; Gimenez et al., 2019; Greer & Weedon, 2012). However, they are designed to record a single value per field of view (FOV), which makes them poorly suited for measuring individual tree canopies and addressing temperature heterogeneity within a single canopy, which is most likely caused by difference in light absorption depending on the position and angles of individual leaves. A limited number of recent studies relying on continuous long-term operation of TICs have validated  $T_c$  measurements for tree canopies (e.g., Aubrecht et al., 2016; Kim et al., 2016, 2018) and proved the potential of TICs to improve our understanding of the interaction between plant physiology and environmental changes.

Leveraging the recent advancement of TIC technology along with synchronous measurements of sap flux, the aim of this study is (1) to understand how the canopy temperature ( $T_c$ ) and VPD near the canopy surface (i.e., at the canopy-air interface;  $VPD_c$ ) are coordinated with species-specific stomatal regulation and (2) to evaluate heterogeneity of  $T_c$  within canopy. Our study species are three canopy-dominant species growing in a temperate forest located in the eastern United States—white oak (*Quercus alba* L.), southern red oak (*Quercus falcata* Michx.), and Virginia pine (*Pinus virginiana* Mill.). Specifically, (1) we first identify hydraulic traits of study species by assessing their stomatal responses under different soil moisture conditions, (2) characterize the diurnal pattern of  $T_c$  and its deviation from  $T_a$  (i.e.,  $T_c - T_a = \Delta T_{c-a}$ ), as well as the corresponding pattern of VPD (i.e.,  $VPD_c - VPD_a = \Delta VPD_{c-a}$ , where  $VPD_a$  is atmospheric VPD), and (3) quantify the sensitivity of  $\Delta T_{c-a}$  and  $\Delta VPD_{c-a}$  to changing environmental conditions for each species in terms of  $T_a$  and  $VPD_a$ . We then discuss how the sensitivities of  $\Delta T_{c-a}$  and  $\Delta VPD_{c-a}$  are associated with species-specific hydraulic traits. We apply these approaches to two different groups of canopy portions divided by their temperatures—defined as sunlit and shaded canopy portions—to further assess heterogeneity of  $T_c$  (sunlit  $T_c - \text{shaded } T_c = \Delta T_{\text{su-sh}}$ ) within a single canopy and compare it across species and see if the influence of the evaporative cooling effect would hold for leaves receiving different levels of incoming radiation.

We hypothesize that species that tend to prioritize water conservation by closing stomates at an earlier level of water stress (i.e., isohydric species) will overall have higher canopy temperatures and be associated with lower sensitivity of  $\Delta T_{c-a}$  and  $\Delta VPD_{c-a}$  to changing moisture conditions. On the other hand, species positioned on the other side of hydraulic spectrum (i.e., prioritizing carbon uptake even under water stress by sustaining stomatal openings; anisohydric) will have cooler surfaces and higher sensitivities. We also

hypothesize that pine will have more homogeneous  $T_c$  distribution within the canopy (i.e., smaller  $\Delta T_{su-sh}$ ) than oaks. This is based on a speculation that pine will allow higher canopy transmittance of solar radiation due to its more heterogeneously distributed needles within the canopy than broadleaves (Cescatti, 1998); consequently, pine needles are expected to warm up more evenly during the day.

## 2. Materials and Methods

### 2.1. Site Description

The study site (Virginia Forest Research Facility) is located in the footprint of an eddy covariance tower running in central Virginia, USA (37° 55'N 78° 16'W). Mean annual temperature and precipitation are 14.0°C and 1,210 mm (over 90% as rain), respectively (1981–2010 normals). The plant functional type is temperate mixed forest dominated by white oak (*Quercus alba* L.), Virginia pine (*Pinus virginiana* Mill.), southern red oak (*Q. falcata* Michx.), red maple (*Acer rubrum* L.), and tulip poplar (*Liriodendron tulipifera* L.). Their relative dominances (=basal area of a species/basal area of all trees  $\times$  100%) within a 500 m radius from the flux tower were 23.6%, 20.0%, 11.9%, 11.5%, and 10.3%, respectively (Chan, 2011). The range of diameter at breast height (DBH) was 2.5 to 81.0 cm, with tree size of second and third quartiles ranging from 4.0 to 15.1 cm. Among the species, Virginia pine (hereafter PV, DBH = 46 cm), white oak (hereafter QA, DBH = 36 cm), and southern red oak (hereafter QF, DBH = 30 cm) were selected as observation targets (one tree for each species) considering their relative abundance, and the position and FOV of the TIC. The heights of the target trees were approximately 25 m. The study period was the growing season of 2019, from mid-July to early October. The soil moisture condition was relatively moist throughout the study period; the volumetric water content (VWC) ranged from 0.25 to 0.45  $\text{cm}^3 \text{cm}^{-3}$ , and the mean value ( $\pm$ standard deviation) was 0.32 ( $\pm$ 0.04)  $\text{cm}^3 \text{cm}^{-3}$  (see Figure S1 in the supporting information for time series and histogram of VWC).

### 2.2. Canopy Temperature Measurements Using a Thermal Imaging Camera

A TIC (FLIR A655sc with 45° FOV lens; FLIR Systems, Wilsonville, OR, USA) was installed on the top platform of the flux tower (40 m tall) to monitor temperature of target canopies (see Figure S2 in the supporting information for preliminary outdoor test results before installation). The distance between the TIC and target canopies was approximately 15 m (Figure 1). To protect the TIC from damage and foreign materials and to maintain optimal conditions for operation, the camera was housed in a thermostatically controlled enclosure (S-Type COOLDOME enclosure; Dotworkz Systems, San Diego, CA, USA). Still images of infrared radiation (spectral range: 7.5–14  $\mu\text{m}$ ) emitted from the target tree canopies were taken every 10 min throughout the study period. The resolution of the images is 640  $\times$  480 pixels. A single output image (file format: SEQ) stores information of thermal signals for all pixels (640  $\times$  480 = 307,200 pixels total) received by the camera sensor and metadata required for thermal signal correction.

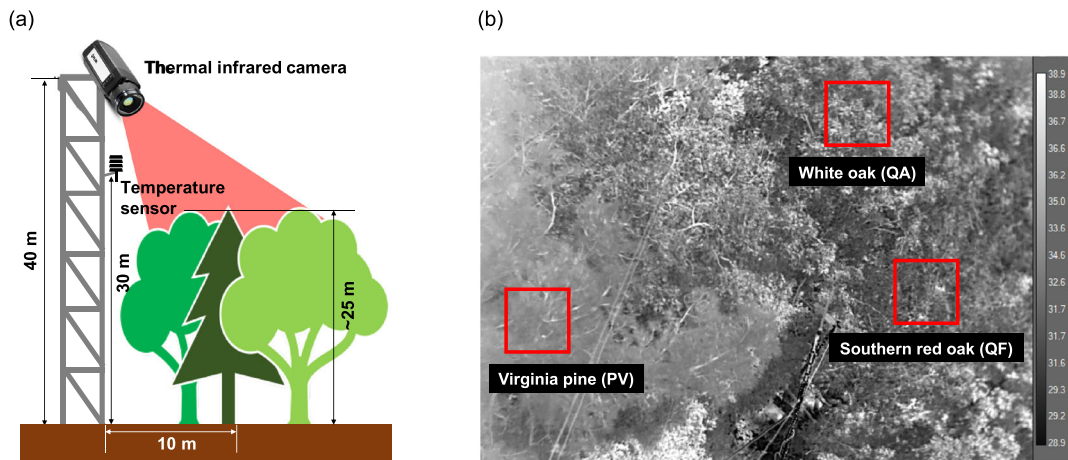
In order to account for atmospheric attenuation of the thermal signal transmitted from the canopies to the camera sensor, we applied the correction suggested by Aubrecht et al. (2016). Here, we only report details about the correction that is unique for our study (see Aubrecht et al., 2016, for more details). Among the parameters required for the correction, sky temperature is measured by a net radiometer installed on the top platform of the tower (CNR4; Kipp & Zonen, Delft, Netherlands). Distance between canopy and camera (=15 m) was estimated based on the dimensions of components in our measurement setup (Figure 1a), including distance of TIC from the ground, tree height, and horizontal distance between tower and trees. Correction was applied for every pixel in the thermal images.

As a proxy of ambient dryness imposed on the canopy, VPD at the canopy surface ( $\text{VPD}_c$ ) was calculated using  $T_c$ . The series of equations for calculating atmospheric VPD ( $\text{VPD}_a$ ) is as follows (Equations 1–3):

$$\text{VPD} = v_{p_{sat}} - v_{p_{air}} \quad (1)$$

$$v_{p_{sat}} = \frac{610.7 \cdot 10^{\frac{7.5 T_a}{237.3 + T_a}}}{1000} \quad (2)$$

$$v_{p_{air}} = \frac{610.7 \cdot 10^{\frac{7.5 T_a}{237.3 + T_a}}}{1000} \cdot \frac{RH}{100} \quad (3)$$



**Figure 1.** Measurement setup (a) and a sample thermal image taken at 13:00 EST on 8 August 2019 (b). Overlaid red boxes ( $70 \times 70$  pixels for each) in the sample thermal image represent regions of interest (ROIs) for different canopies. The spatial resolution is  $\sim 2.5$  cm. The scale bar in panel (b) indicates surface temperature distribution (unit:  $^{\circ}\text{C}$ ).

where  $vp_{\text{sat}}$  is saturated vapor pressure of the air,  $vp_{\text{air}}$  is actual vapor pressure of the air, and  $RH$  is relative humidity. The  $T_a$  and  $RH$  were measured by a temperature and humidity probe (HMP45; Vaisala, Helsinki, Finland) installed on the tower at the height of 30 m from the ground. To calculate  $VPD_c$ , the  $T_a$  in the equation for  $vp_{\text{sat}}$  (Equation 2) was replaced with  $T_c$ , such that

$$vp_{\text{sat, canopy}} = \frac{610.7 \cdot 10^{\frac{7.5 T_c}{237.3 + T_c}}}{1000} \quad (4)$$

The same  $RH$  was used to calculate both  $VPD_a$  and  $VPD_c$  with the assumption that the differences in mixing of water vapors at the leaf surface (for  $VPD_c$ ) and in the air near the leaf surface (for  $VPD_a$ ) were negligible.

Finally, the representative sunlit and shaded  $T_c$  for each species were determined by setting a square sample area (i.e., region of interest, ROI) for each canopy (Figure 1). Our definitions of sunlit and shaded portions of a canopy are based on the assumptions that leaves are mainly warmed by light absorption and difference in transpiration rate among the leaves within ROI is negligible. The size of each ROI was  $70 \times 70$  pixels. The ROIs were chosen by visual inspection of images, ensuring pixels of bare ground are not included. The sunlit and shaded  $T_c$  were determined in a conservative manner to avoid selecting the portions of extreme  $T_c$  only. We assumed that the pixels within the highest 25% of temperatures represent the canopy portions absorbing ample sunlight;  $T_c$  from these pixels were averaged to define “sunlit  $T_c$ .” On the other hand, the pixels within the lowest 50% temperatures were assumed to be  $T_c$  of the canopy portions absorbing significantly less sunlight than the sunlit portions;  $T_c$  from these pixels were averaged to define “shaded  $T_c$ .” For these definitions, data collected during daytime from 9 a.m. to 4 p.m. were used. Both sunlit and shaded  $T_c$  were converted into hourly data by taking averages of the original data collected every 10 min.

### 2.3. Sap Flux Measurements

Sap flux densities ( $J_s$ ,  $\text{cm}^3 \text{ cm}^{-2} \text{ hr}^{-1}$ ) of the target trees were monitored using a measurement system (HRM30 Sap Flow Sensors and SL5 Smart Logger; ICT International, Armidale, New South Wales, Australia) that adopts the heat ratio method (Burgess et al., 2001). The sap flux sensors consist of three 3.5-cm-long needles inserted into the tree trunk at breast height in a radial direction, with a vertical separation distance of 5 mm; a pair of temperature sensors detect temperature variation and a heater positioned between two sensors generates heat pulses every 10 min. The  $J_s$  was measured at a radial depth of 12.5 mm from the epoxy base of the sensors, where thermocouples are positioned, to represent  $J_s$  at sapwood area. Correct radial placement of thermocouples was inferred from the sapwood depth of each tree, which



were measured after collecting tree-ring cores. The  $J_s$  was originally recorded every 10 min and was converted into hourly data by taking averages of the original data for each species.

To identify hydraulic traits of the study species, we used a diagnostic framework (Equation 7) that combines the concept of soil-plant-atmosphere continuum (SPAC; Equation 5) and a simplified relationship between  $J_s$  and  $g_s$  (Oren et al., 1999; Equation 6) as follows:

$$J_s(\text{as a proxy of transpiration}) = K(\Psi_s - \Psi_L) \quad (5)$$

$$J_s = g_s \times \text{VPD}_a \quad (6)$$

$$g_s = \frac{K(\Psi_s - \Psi_L)}{\text{VPD}_a} = \frac{K(\Delta\Psi)}{\text{VPD}_a} \quad (7)$$

where  $K$  is whole-plant hydraulic conductance,  $\Psi_s$  is soil water potential,  $\Psi_L$  is leaf water potential, and  $\Delta\Psi$  is the difference between  $\Psi_s$  and  $\Psi_L$ . The contribution of gravitational head loss to  $\Delta\Psi$ , which is relatively constant over weekly to monthly time scales, is neglected. The equation suggests that the tendency for plants to regulate leaf water potential ( $\Psi_L$ ) under contrasting soil moisture conditions ( $\Psi_s$ ) is reflected by stomatal regulation ( $g_s$ ) and its response to changing  $\text{VPD}_a$ . Species that tend to conserve water under low  $\Psi_s$  by keeping  $\Psi_L$  constant (thus rapid decrease in  $\Delta\Psi$ ) would show larger decreases in  $g_s$  when soil moisture is generally low (i.e., low  $\Psi_s$ ), compared to the species that are less conservative in their regulation of  $\Psi_L$ . Specifically, we compared  $g_s$ - $\text{VPD}_a$  (or  $\text{VPD}_c$ ) relationships under two contrasting soil moisture conditions; wet ( $\text{VWC} \geq 0.28 \text{ cm}^3 \text{ cm}^{-3}$  within the depth of 30 cm) and dry ( $\text{VWC} < 0.28 \text{ cm}^3 \text{ cm}^{-3}$ ) conditions. Our assumption is that VWC represents the amount of soil moisture trees can access, and the study trees have the same accessibility to the soil moisture (see section 4.1 for limitations of the assumption). The threshold of VWC to define wet and dry conditions was decided after investigating the distribution of VWC during the study period (see Figure S1 in the supporting information for histogram of VWC). Specifically, we set two criteria to determine dry condition: (1) Dry condition is less common condition compared to moderate to wet condition in our site (given  $\text{VWC} > 0.25 \text{ cm}^3 \text{ cm}^{-3}$  throughout the study period), and thus, we assume dry condition should not exceed 30% of entire period; (2) dry condition must have at least 100 sampling numbers to avoid uncertainty arising from having low sample numbers. As a result, 28% of data were classified as dry condition and the rest 72% as wet condition. To facilitate comparison of stomatal responses among the study species, the estimated  $g_s$  was normalized by dividing  $g_s$ - $\text{VPD}_a$  curves by the value of  $g_s$  at  $\text{VPD}_a = 1 \text{ kPa}$ .

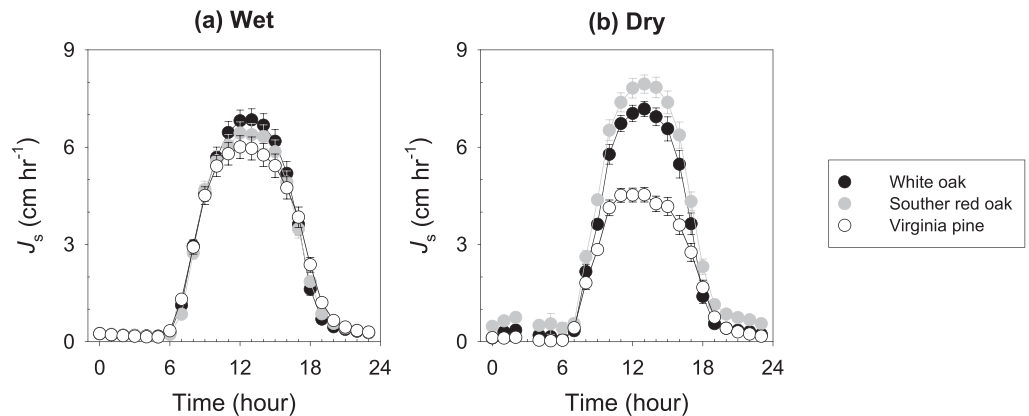
## 2.4. Data Analyses

The species-specific physiological responses (i.e., normalized  $g_s$ ,  $\Delta T_{c-a}$  ( $=T_c - T_a$ ),  $\Delta \text{VPD}_{c-a}$  ( $=\text{VPD}_c - \text{VPD}_a$ ), and  $J_s$ ) to changing environmental conditions (i.e.,  $T_a$  and  $\text{VPD}_a$ ) were assessed based on the data collected during daytime from 9 a.m. to 4 p.m. To clarify the patterns of these relationships, the physiological data were partitioned into discrete bins spanning a range of environmental drivers (e.g., in Figures 5 and 6, each symbol represents a mean value of data points belong to each bin). The widths of the data range for individual bins are not necessarily identical because the size was determined based on the data distribution; every bin was forced to have an equal number of samples (approximately 80) in order to prevent a bias arising from skewed data distribution (e.g., the sample mean from the extreme ends of  $T_a$  or  $\text{VPD}_a$  may not represent the population mean if the sample values are spread out over a wide range and the sample size is small). Difference in  $\Delta T_{c-a}$  or  $\Delta \text{VPD}_{c-a}$  between sunlit and shaded canopy portions (in Figures 5 and 6) was evaluated by using one-way analysis of variance and Tukey post hoc test at a significance level of 0.05 using SPSS Statistics (v.27). The diurnal patterns of plant physiological responses were represented by mean hourly data over the study period.

## 3. Results

### 3.1. Species-Specific Hydraulic Traits

All species had a unimodal diurnal pattern of  $J_s$  (Figure 2; see Figure S3 in the supporting information for the time series of  $J_s$  variation over the entire observation period), linked to the diurnal range of photosynthetically active radiation. The  $J_s$  was minimal until early morning, increased until midday, and then decreased gradually, reaching minimal  $J_s$  again by sunset. There was a species-specific difference, however, when  $J_s$



**Figure 2.** Mean diurnal patterns of species-specific sap flux density ( $J_s$ ) under wet (volumetric water content [VWC]  $\geq 0.28 \text{ cm}^3 \text{ cm}^{-3}$ , a) and dry (VWC  $< 0.28 \text{ cm}^3 \text{ cm}^{-3}$ , b) conditions (QA: white oak, QF: Southern red oak, and PV: Virginia pine). Error bars represent standard errors of the means (95% confidence).

under different soil moisture (VWC) conditions were compared. While the  $J_s$  of PV was substantially decreased under drier conditions ( $\sim 25\%$  around noon), oaks increased  $J_s$  ( $\sim 5\%$  and  $\sim 25\%$  around noon for QA and QF, respectively).

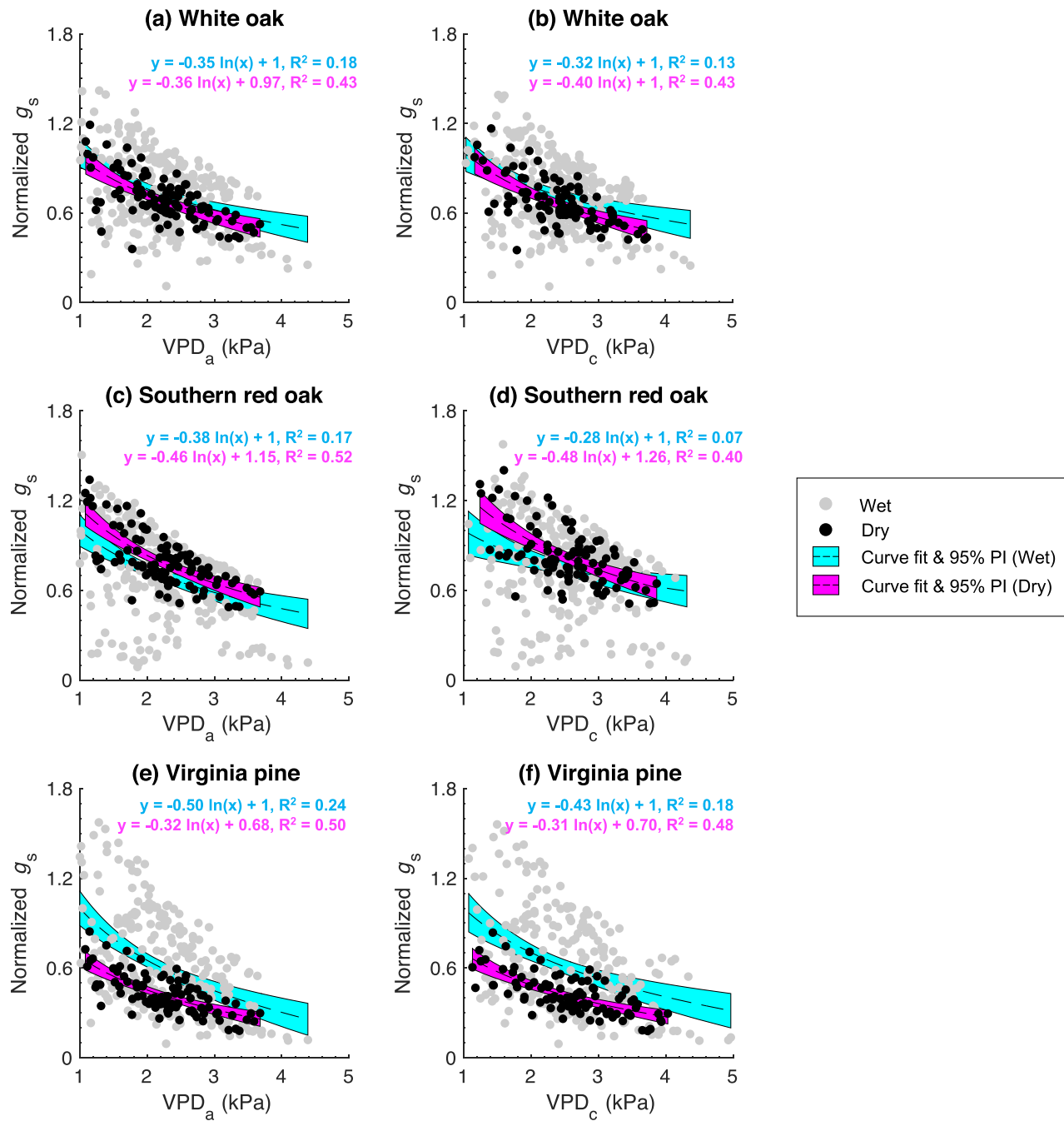
Species-specific differences in stomatal regulation were also evident when the response of  $g_s$  to changing VPD was evaluated under contrasting soil moisture conditions (Figure 3). Specifically,  $g_s$  of PV under dry conditions was noticeably lower than under wet conditions (Figures 3e and 3f). On the other hand,  $g_s$  of QA were very similar in both wet and dry periods (Figures 3a and 3b), while QF had even higher  $g_s$  during drier period (Figures 3c and 3d). In addition,  $g_s$  response to VPD under wet condition was more sensitive for PV than for other species. Sensitivity of  $g_s$  to changing VPD<sub>c</sub> (Figures 3b, 3d, and 3f) was not substantially different from the sensitivity to changing VPD<sub>a</sub> (Figures 3a, 3c, and 3e), although the range of VPD<sub>c</sub> was slightly different from VPD<sub>a</sub> which was more prominent for PV: the maximum VPD<sub>a</sub> and VPD<sub>c</sub> for PV were 4.39 and 4.97 kPa, respectively (Figure 3e vs. Figure 3f).

### 3.2. Diurnal Patterns of $T_c$ and VPD<sub>c</sub>

Mean diurnal patterns of  $T_c$  were similar to the pattern of  $T_a$  for all species, reaching a minimum temperature around 5 a.m. and maximum around 1 p.m. (Figure 4a; see Figure S4 in the supporting information for  $T_a$  and  $T_c$  variation over entire observation period). For all species, sunlit  $T_c$  was higher than  $T_a$  during most of the daytime (from 8 a.m. to 5 p.m.; Figure 4b), up to  $0.9^\circ\text{C}$  for QA (peaked at 10 a.m.),  $1.6^\circ\text{C}$  for QF (peaked at noon), and  $1.5^\circ\text{C}$  for PV (peaked at 3 p.m.). Among the species, PV's  $T_c$  warmed more slowly in the morning and cooled more slowly in the afternoon (Figure 4b). Specifically, in the morning, QA (from 7 a.m. to 10 a.m.) and QF (from 7 a.m. to 1 p.m.) had higher sunlit  $T_c$  than PV up to  $0.9^\circ\text{C}$  and  $0.8^\circ\text{C}$ , respectively (both peaked at 8 a.m.). In the afternoon, sunlit  $T_c$  of PV was higher than the other species—up to  $1.3^\circ\text{C}$  higher than both QA (peaked at 5 p.m.), and QF (peaked at 6 p.m.). There was slight increase in  $\Delta T_{c-a}$  during night (from 9 p.m. to 4 a.m.) across all species, indicating slower cooling of  $T_c$  than  $T_a$  at night.

Compared to the sunlit  $T_c$ , the diurnal variations of the shaded  $T_c$  were notably larger across the species (Figure 4c). PV showed lower diurnal variation of shaded  $T_c$  than the other species; diurnal variation of the difference between sunlit and shaded  $T_c$  was  $0.2\text{--}1.5^\circ\text{C}$  for QA,  $0.2\text{--}1.9^\circ\text{C}$  for QF, and  $0.3\text{--}0.9^\circ\text{C}$  for PV. The difference was highest at 1 p.m. Shaded  $T_c$  of QA and QF, unlike the sunlit  $T_c$ , were always lower than  $T_a$  (up to  $0.9^\circ\text{C}$  for QA and  $1.2^\circ\text{C}$  for QF, peaked at 7 a.m. and 6 p.m.) throughout the day. On the other hand, PV maintained higher  $T_c$  than  $T_a$  (up to  $0.7^\circ\text{C}$ , peaked at 5 p.m.) at most times except a few hours in the morning (from 6 a.m. to 9 a.m.).

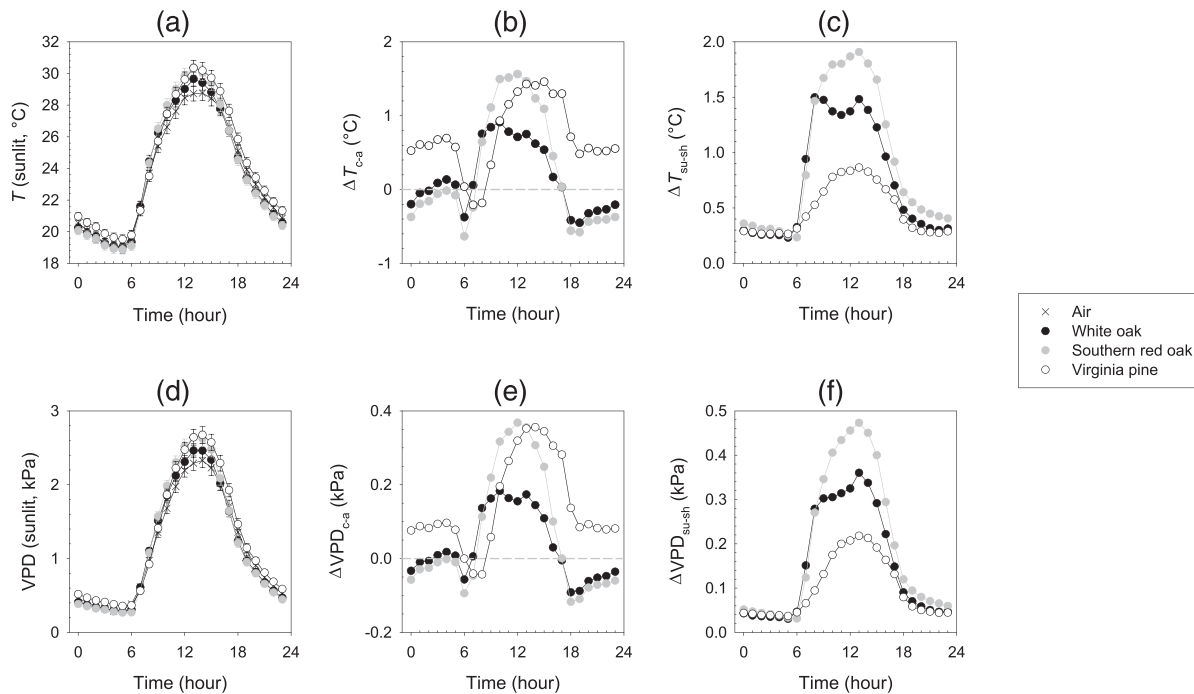
The diurnal patterns of VPD<sub>c</sub> for the three species closely followed the patterns of corresponding  $T_c$  (Figure 4d). All species had higher sunlit VPD<sub>c</sub> than VPD<sub>a</sub> during most of the daytime (between 9 a.m. and 5 p.m.), up to  $0.2 \text{ kPa}$  for QA (peaked at 10 a.m.),  $0.4 \text{ kPa}$  for QF (peaked at noon), and  $0.4 \text{ kPa}$  for



**Figure 3.** Species-specific relationships between normalized stomatal conductance ( $g_s$ ) and VPD<sub>a</sub> (a, c, e) or VPD<sub>c</sub> (b, d, f) under contrasting soil moisture conditions. The  $g_s$  values under both wet and dry conditions were normalized by dividing  $g_s$  by wet condition  $g_s$  at VPD<sub>a</sub> or VPD<sub>c</sub> = 1 kPa. Solid curves denote curve fits for non-linear regression models of normalized  $g_s$  – VPD (equation: normalized  $g_s = a \times \ln(\text{VPD}) + b$ ). Dashed curves denote 95% prediction intervals of the regression models.

PV (peaked at 2 p.m.; Figure 4e). Among the species, PV usually had higher sunlit VPD<sub>c</sub> than the other species—up to 0.3 kPa higher than both QA and QF (peaked at 5 p.m.)—except in the morning, where both QA (from 7 a.m. to 9 a.m.) and QF (from 8 a.m. to 1 p.m.) showed higher VPD<sub>c</sub> than PV up to 0.2 kPa (both peaked at 8 a.m.).

Compared to the sunlit canopy portions, shaded portions always had lower VPD<sub>c</sub>; the range of difference between sunlit VPD<sub>c</sub> and shaded VPD<sub>c</sub> during a day was 0.0–0.4 kPa for QA, 0.0–0.5 kPa for QF, and



**Figure 4.** Mean diurnal variations of temperature (a) and VPD (d), the corresponding differences between sunlit canopy and air (b and e), and sunlit and shaded canopy portions (c and f) measurements for each species. Error bars represent standard errors of the means of hourly data (95% confidence). Horizontal dashed lines denote reference lines for  $\Delta T = 0$  or  $\Delta VPD = 0$ .

0.0–0.2 kPa for PV (Figure 4f). The difference was highest at 1 p.m.  $VPD_c$  for the shaded portions of QA and QF was always lower than  $VPD_a$  (up to 0.2 kPa for both species, peaking at both 7 a.m. and 6 p.m.) throughout the day. On the other hand, PV maintained higher  $VPD_c$  for the shaded portions than  $VPD_a$  at most times (up to 0.2 kPa, peaked at 2 p.m.), except a few hours in the morning (from 6 a.m. to 9 a.m.).

### 3.3. Species-Specific Response of $T_c$ and $VPD_c$ to Changing Condition

The response of  $\Delta T_{c-a}$  to changing  $T_a$  was different across species (Figures 5a–5c). Specifically, PV maintained almost constant  $\Delta T_{c-a}$  while the oak species had decreasing  $\Delta T_{c-a}$  with rising  $T_a$ . The differences among the species were consistent in both sunlit and shaded canopy portions, with significantly lower  $\Delta T_{c-a}$  for shaded portions than sunlit portions across the range of  $T_a$  ( $p < 0.05$ ).

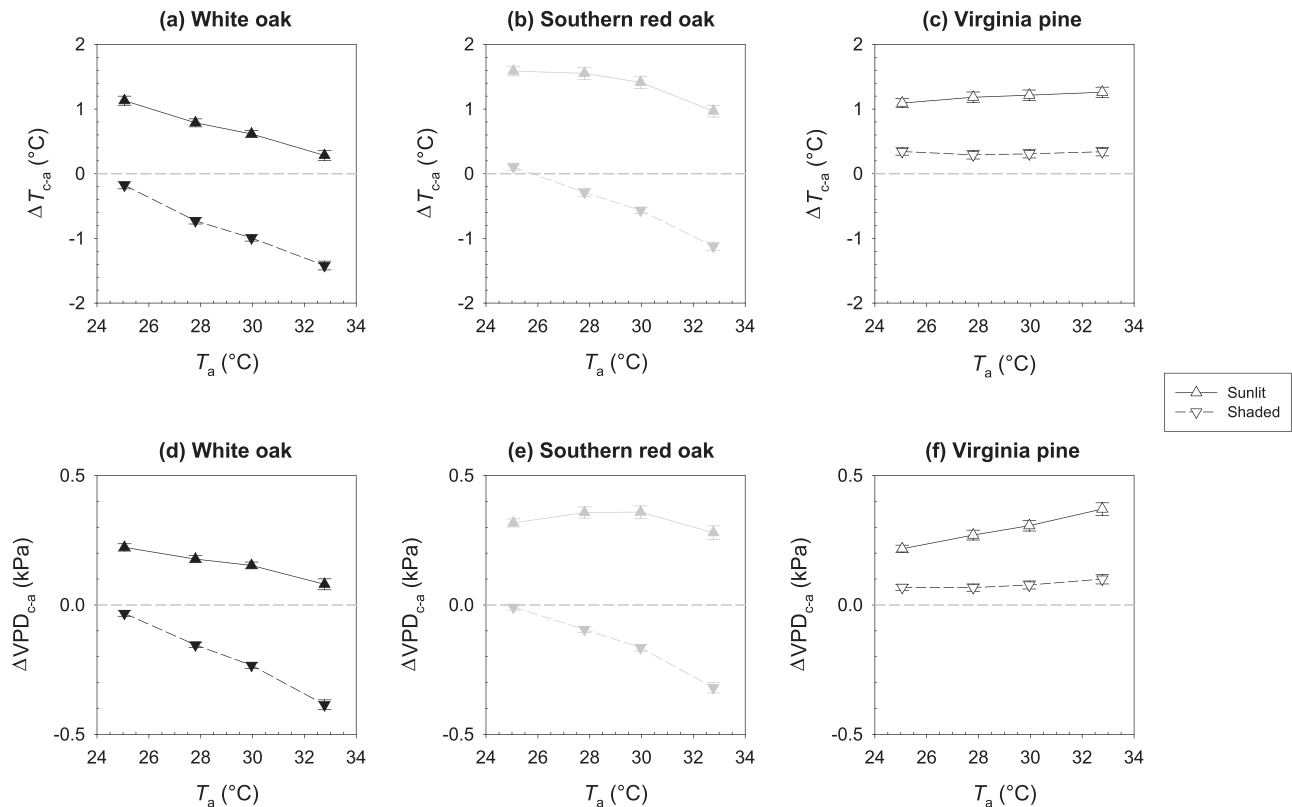
The differences among the species were more evident when the response of  $\Delta VPD_{c-a}$  to changing  $T_a$  was compared (Figures 5d–5f). In the case of sunlit canopy portions, QA consistently decreased  $\Delta VPD_{c-a}$  with rising  $T_a$ , while PV consistently increased  $\Delta VPD_{c-a}$ . The response of QF to changing  $T_a$  was more or less constant. In the case of shaded portions, both QA and QF consistently decreased  $\Delta VPD_{c-a}$  with rising  $T_a$ , while the  $\Delta VPD_{c-a}$  of PV remained constant.

The species-specific differences were also apparent from the responses of  $\Delta T_{c-a}$  and  $\Delta VPD_{c-a}$  to changing  $VPD_a$  (Figure 6). The differences among the species appears to be associated with species-specific stomatal regulation, and consequently, transpirational response (Figures 6g–6i); PV reduced  $J_s$  earlier at lower  $VPD_a$  than oak species, and leaf cooling did not completely offset leaf warming as shown in  $\Delta T_{c-a}$  (Figures 6a–6c) and  $\Delta VPD_{c-a}$  responses (Figures 6d–6f).

## 4. Discussion

We investigated (1) species-specific responses of canopy temperature to changing air temperature or atmospheric dryness, and their associations with hydraulic traits, and (2) heterogeneities in canopy temperature across species and within a single canopy. The analysis leveraged continuous measurements using a TIC, which has not been widely applied for studies of tree canopies until recently (see Aubrecht et al., 2016,





**Figure 5.**  $\Delta T_{c-a}$  ( $=T_c - T_a$ ; a–c) and  $\Delta VPD_{c-a}$  ( $=VPD_c - VPD_a$ ; d–f) as functions of  $T_a$ . Note that the result is based on data collected during the daytime from 9 a.m. to 4 p.m. Each point represents mean value of samples from each bin of  $T_a$ . Error bars represent standard errors of the means (95% confidence). Horizontal dashed lines denote reference lines for  $\Delta T_{c-a} = 0$  or  $\Delta VPD_{c-a} = 0$ .

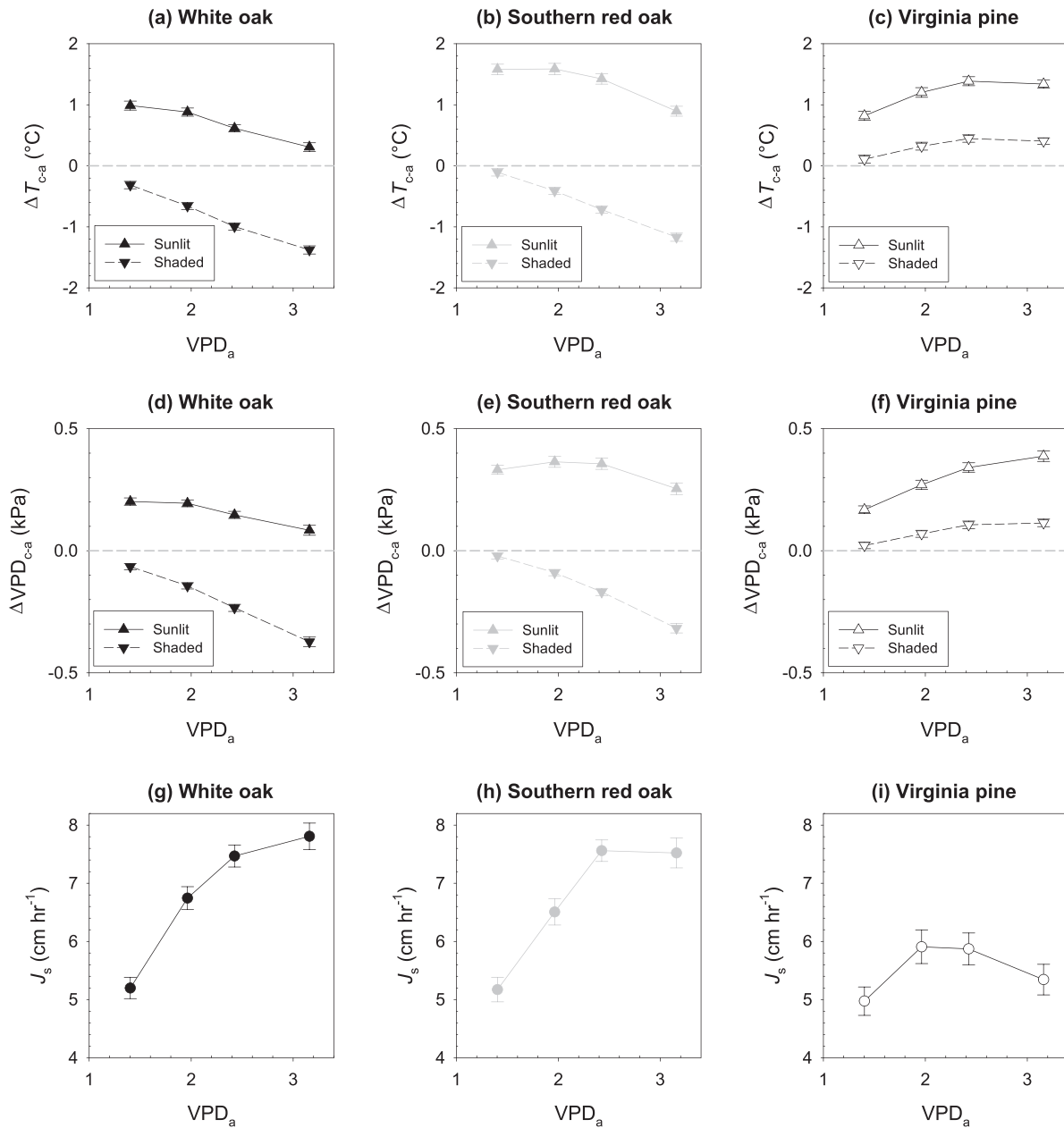
and Kim et al., 2016, 2018, for earlier work). In the following sections, we discuss more details about a link between canopy temperature variability and plant hydraulic traits, its implications, and application of TICs.

#### 4.1. Link Between Sensitivity in Canopy Temperature and Hydraulic Traits

One of our goals of the study was to understand how the  $T_c$  and VPD near the canopy surface are coordinated with species-specific stomatal regulation, that is, isohydricity. We found that  $T_c$  was notably higher than  $T_a$  for all observed species during the daytime, ranging from 1.0°C to 1.6°C (Figure 4b). The canopy was substantially cooled (relatively to  $T_a$ ) under high  $T_a$  (Figure 5) and  $VPD_a$  (Figure 6) in white oak (QA) and southern red oak (QF) but less so in Virginia pine (PV), (Figure 4b). Moreover, it appears that the lower cooling effect of PV during the daytime persists into the nighttime (Figure 4b), imposing a higher heat load for most of the day, when compared to the other species. While it is not our research focus, understanding what limits net carbon uptake at high temperatures will become increasingly important since we are anticipating warmer climate in the near future.

The difference in  $T_c$  sensitivity among the species appears to be associated with hydraulic traits affecting stomatal regulation, and hence transpiration, in response to changing environmental conditions. Our diagnosis of species-specific hydraulic traits suggests that PV regulates stomatal conductance ( $g_s$ ) more actively in response to changing moisture conditions (both soil moisture and atmospheric dryness) than QA and QF (Figure 3). Our identification of hydraulic traits agrees with previous studies identifying oak species as more anisohydric (Roman et al., 2015; Yi et al., 2019) and pine species as more isohydric species (Burkhardt & Pariyar, 2016; Klein et al., 2011).

The response of transpiration (represented by sap flux density,  $J_s$ ) to changing moisture conditions was caused by stomatal regulation. The  $J_s$  of oak species persistently increased with rising  $VPD_a$  (Figures 6g and 6h) and decreasing soil moisture (Figure 2), due to their sustained  $g_s$  (note  $J_s \propto g_s \cdot VPD$  given tree



**Figure 6.**  $\Delta T_{c-a}$  ( $=T_c - T_a$ ; a-c),  $\Delta VPD_{c-a}$  ( $=VPD_c - VPD_a$ ; d-f), and  $J_s$  (g-i) as functions of  $VPD_a$ . Note the result is based on data collected during the daytime from 9 a.m. to 4 p.m. Each point represents mean value of samples from each bin of  $T_a$ . Error bars represent standard errors of the means (95% confidence). Horizontal dashed lines denote reference lines for  $\Delta T_{c-a} = 0$  or  $\Delta VPD_{c-a} = 0$ .

hydraulic conductivity and soil moisture are not limiting). Due to the higher cooling effect of oak species driven by higher transpiration rate under higher  $VPD_a$ , their heat ( $\Delta T$ ) and moisture ( $\Delta VPD$ ) stresses were reduced with increasing  $VPD_a$ . On the other hand, the  $J_s$  of PV saturated at a lower  $VPD_a$  (Figure 6i) and decreased when soil moisture was lower (Figure 2b), resulting in a less effective cooling effect than the oak species (e.g., sustained  $\Delta T$  and increased  $\Delta VPD$  with rising  $VPD_a$ ; Figure 6). Overall, our results suggest that the oak species perform better in reducing the stresses from both increasing heat and dryness compared to the PV.

Meanwhile, it is important to consider the difference in rooting depth and soil moisture access among the species when diagnosing hydraulic traits. For instance, higher  $g_s$  of the oak species than the pine under

drier condition (Figure 3) can be attributed to their deeper rooting depth, which allows better access to moisture in the deeper soil when surface water is insufficient (Hochberg et al., 2018). Nonetheless, the impact of different soil moisture access among the species, if any, on our characterization of hydraulic traits is likely minimal. Throughout the study period, the VWC was fairly high without dropping below  $0.25 \text{ m}^3 \text{ m}^{-3}$  within the soil depth of 30 cm. Moreover, our definition of dry soil conditions was relatively conservative considering almost 30% of cases were assumed to be dry. Despite the conservative definition, stomatal regulation of pine tree was clearly different from that of oak species. Therefore, we presume the difference in rooting depth and soil moisture access has little influence on the identification of hydraulic trait for the study species, at least for the relatively normal soil moisture regimes experienced during this study.

It was noteworthy that while using  $\text{VPD}_c$  (i.e., the VPD from a plant perspective) did not change our identification of hydraulic traits of the study species,  $\text{VPD}_c$  was significantly higher than  $\text{VPD}_a$  during the daytime (Figure 4e). Moreover, the sensitivity of  $g_s$  to VPD (i.e., the slope of  $g_s$  - VPD; Figure 3) was slightly overestimated when  $\text{VPD}_a$  was used instead of  $\text{VPD}_c$ , especially for the species that showed relatively large differences between  $T_c$  and  $T_a$  (QF and PV; Figure 4b). The difference in the slope was driven by the wider range of  $\text{VPD}_c$  than  $\text{VPD}_a$ , making the  $g_s$ - $\text{VPD}_a$  slope lower (i.e., more negative) than the  $g_s$ - $\text{VPD}_c$  slope (Figure 3). This implies that the  $g_s$  sensitivity to VPD is more likely to be overestimated when using  $\text{VPD}_a$  instead of  $\text{VPD}_c$  for more isohydric species like PV.

Our assumption for the difference between estimations of  $\text{VPD}_a$  and  $\text{VPD}_c$  was due to the choice of different temperature metrics ( $T_a$  vs.  $T_c$ ) to calculate saturated vapor pressure of the air ( $vp_{\text{sat}}$  vs.  $vp_{\text{sat,canopy}}$ , Equations 2 and 4). However, one should note that VPD changes with  $RH$ , since actual vapor pressure of the air ( $vp_{\text{air}}$ ; Equation 3) is a function of  $T_a$  and  $RH$ . While our  $T_a$  and  $RH$  were measured at a height similar to the tree height (Figure 1a), measured  $RH$  can be lower than actual  $RH$  at immediate proximity to the leaf surface due to better turbulent mixing of air, which can consequently result in smaller  $\text{VPD}_c$  than our estimation. Similarly, measured  $T_a$  can be different than  $T_a$  at immediate proximity to the leaf surface due to turbulent mixing of air and temperature gradient along the atmospheric layers. Although we assumed these micro-meteorological influences to be negligible, further leaf-level studies coupled with aerodynamics can help reduce the uncertainties we have discussed.

#### 4.2. Other Factors Influencing Canopy Temperature Sensitivity

$T_c$  can be also influenced by physical attributes of the leaves or canopy (leaf morphology or canopy structure). For instance, the large surface area per leaf of the oaks may allow their individual leaves to intercept more solar radiation than the pine needles do, resulting in their faster  $T_c$  increase in the morning (Figure 4b) when the difference in evaporative cooling effects among the species was relatively small (Figure 2). Meanwhile, lower canopy albedo for the pine canopy due to the darker hue of its needles (Juang et al., 2007) may contribute more positively to the  $T_c$  by absorbing and retaining heat better than the other species. This might have contributed to the high  $T_c$  of the pine tree in the afternoon and the retained high  $T_c$  even at night (Figure 4b). The different leaf warming effects among the species work independently from evaporative cooling effect, which complicates assessment of leaf thermoregulation: This is a major reason why we tested the relationship between evaporative cooling effect and canopy temperature indirectly by grouping data into different  $\text{VPD}_a$  (Figure 6), since VPD is directly associated with the leaf cooling mechanism (i.e.,  $g_s$  regulation and  $J_s$  variability as a result) but not with warming effects. Therefore, comprehensive study including both warming and cooling factors together is required to clarify their impact on the leaf thermoregulation and its physiological consequence.

Additionally, more homogeneous  $T_c$  distribution of the pine (Figures 4c and 4f) may be attributable to its leaf morphology and foliage clumping, as these attributes result in more heterogeneous spatial distribution of leaf area within the canopy than typical broad-leaved trees (Cescatti, 1998). The heterogeneous distribution of leaf area creates more gaps that allows higher canopy transmittance of solar radiation (Cescatti, 1998; Duursma & Mäkelä, 2007; Fotis & Curtis, 2017), making  $T_c$  distribution more homogenous within canopy. More homogenous spatial distribution of leaf area of the oak species, on the other hand, may cause the larger difference between sunlit and shaded  $T_c$  by intercepting solar radiation before it enters deeper into the canopy.

### 4.3. Implications of Variability in Cooling Effect Among Different Species

Among our studied species, PV showed large differences between  $T_c$  and  $T_a$ , and its sunlit  $T_c$  was up to 1.3°C higher than the oak species (Figure 4). This difference is notable considering the trees are co-occurring and experiencing almost identical environmental conditions. Furthermore, the difference in  $T_c$  among the species is comparable to the difference of surface temperature among different land covers/uses reported in other studies ( $\sim +1^\circ\text{C}$  in temperate zone by forest cover removal; Alkama & Cescatti, 2016; Bright et al., 2017). There are also relevant studies comparing surface temperature between co-located forests and grasslands in the eastern United States using radiometric measurements; for instance, Novick and Katul (2020) compared surface temperature ( $T_{\text{surf}}$ ) of hardwood forest, pine forest, and grasslands using a radiometric approach and showed that  $T_{\text{surf}}$  of either type of the forests were cooler than the grassland (often exceeding 5°C) during the midday of a growing season. Zhang et al. (2020) reported that  $T_{\text{surf}}$  of grasslands was higher by 1–2°C than co-located forests at annual scale, with growing season temperature difference largely attributable to greater water use by forests.

One should note that, despite the notable difference in  $T_c$  across the co-occurring species, their cooling effect on  $T_c$  is not equal to the cooling effect on climate or  $T_a$ , which is more relevant for addressing climate mitigation roles of different vegetation types or species. While extending  $T_c$  or  $T_{\text{surf}}$  changes regulated by plants to  $T_a$  has been challenging due to the impact of heat transfer mechanism mediated by height of atmospheric boundary layer, Novick and Katul (2020) used a novel approach to assess the impact of reforestation on both  $T_a$  and  $T_{\text{surf}}$ . They showed that although the difference in  $T_a$  between forests and grassland was smaller than their difference in  $T_{\text{surf}}$ ,  $T_a$  at the forests was 2–3°C cooler than the grassland during daytime growing season, which is still substantial.

Overall, despite the smaller cooling effect of vegetation on  $T_a$  than on  $T_c$ , evidence suggest that changes in species composition in forest ecosystems might cause changes in  $T_c$  that is comparable to the change caused by land use/cover change, which is likely to contribute to mitigate or aggravate global warming consequently. Our study species represent typical needleleaf and broadleaf trees that can be found in temperate forests in the eastern United States, and they are also typically known to be positioned very differently on the spectrum of hydraulic traits regarding their stomatal regulation. Specifically, oaks sustain stomatal opening under water stress (anisohydric; Roman et al., 2015; Yi et al., 2019) and pines as species closing stomata at earlier stages of water stress (isohydric; Burkhardt & Pariyar, 2016; Klein et al., 2011). A simple conjecture suggests that regions experiencing a shift in dominant species from anisohydric to isohydric (e.g., declined oak species in the eastern United States; Abrams, 2003; Flatley et al., 2013) might experience a reduction in the local cooling benefit provided by forested ecosystems. However, more work is needed to fully understand how species with contrasting hydraulic traits will respond to the warmer and drier conditions expected in the future. Moreover, vulnerability or mortality of iso/anisohydric species under those conditions is hard to generalize. Therefore, further advancement in on-going modeling efforts to incorporate plant hydraulic traits (e.g., SPAC model; Williams et al., 1996; Xu et al., 2016) into land surface models, along with more comprehensive understanding of plant thermoregulation in terms of energy balance and species-specific hydraulic traits, would inform the usefulness of reforestation as a local climate mitigation and adaptation strategy.

### 4.4. Thermal Imaging Camera as a Tool for Canopy Temperature Monitoring

High-resolution thermal images provide snapshots of the within- and cross-canopy heterogeneity in  $T_c$ . The TIC was ideal for  $T_c$  measurement of tall trees primarily due to its ability to overcome very low accessibility of tall-tree canopies. Unlike the infrared radiometers that record a single value of temperature per device, TIC can record temperatures at different parts of an image at the same time and allows visual inspection if the target was properly recorded, which enables partitioning sunlit and shaded portions within a canopy as shown in our study. Furthermore, being able to measure temperature of multiple positions from a single tree canopy means users can collect replication data for a single canopy at the same moment while all conditions are identically applied for those replications. This helps eliminate (1) biases that can arise otherwise, for instance, data collection at different times of similar conditions (but not exactly the same) to investigate plant-environment interactions, or (2) instrumental biases (e.g., positioning of instrument and synchronization of settings) that can arise from using separate instruments at the same time to measure different targets.

Therefore, the TIC has an advantage over infrared radiometers in recording temperature of targets with highly heterogeneous surfaces, while infrared radiometers should work well for targets with homogeneous surfaces (e.g., cropland).

We used the fixed ROIs to record mean  $T_c$ . The canopies were stationary under the most moderate conditions; however, canopies can sway when wind is high. Although the positions for ROIs were carefully chosen to always properly target the canopies considering the trajectories of canopy movements, developing an algorithm to track the movement of targets and measuring them might improve the results. Nevertheless, our measurements were highly precise with very low variations (e.g., low standard errors throughout the results; also see Figure S2 in the supporting information showing a result from preliminary test performed on multiple leaves of a plant growing in the pot), suggesting the use of fixed targets can still provide reliable measurements if their positions are carefully set.

## 5. Conclusion

We found notable differences among the study species in the sensitivities of canopy temperature ( $T_c$ ) and VPD at the canopy surface ( $VPD_c$ ) to rising air temperature ( $T_a$ ) and atmospheric water demand ( $VPD_a$ ), which was closely related to species-specific hydraulic traits. Specifically, the isohydric species, which regulated stomatal conductance more strongly under conditions of drier soil moisture, experienced larger differences between  $T_c$  and  $T_a$ . It is noteworthy that the variability of sunlit  $T_c$  among the species was up to 1.3°C, which was comparable to the surface temperature variation caused by land use change reported elsewhere (1°C to 2°C). This implies that change in species composition can have a significant impact on the local to regional surface temperature. Meanwhile, we also found the heterogeneity of  $T_c$  within the canopy (from the difference between sunlit and shaded  $T_c$ ) can be remarkably different depending on the species, which may be attributed to their different leaf and canopy structure. Therefore, more precise evaluation of the feedback between plant thermoregulation and climate system would benefit from the improved and comprehensive understanding of plant hydraulic traits and energy exchange processes occurring at both canopy-atmosphere interface and within the canopy.

## Data Availability Statement

Data used in this study are available on Zenodo (<https://doi.org/10.5281/zenodo.3896812>).

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